

Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

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0.1 Summary

- Humans produce higher novelty (surprisal) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- AI contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
# Load Field Data - using standardized column names
df_res_field <- read_csv(here("data/human-ai/processed/novelty_scores.csv")) %>%
  rename(
    user = author_1,
    ai = author_2,
```

```

    user_surprise = author_1_surprise,
    ai_surprise = author_2_surprise,
    user_transience = author_1_transience,
    ai_transience = author_2_transience
  ) %>%
  filter(!is.na(user_surprise) & !is.na(ai_surprise)) %>%
  mutate(
    user_resonance = user_surprise - user_transience,
    ai_resonance = ai_surprise - ai_transience,
    datatype = "Field"
  ) %>%
  drop_na(user, ai) %>%
  filter(conversation_id != "conv_0bb56093-3033-4615-bb70-ebfa4135589a", turn > 1, ai_surprise < 0)

```

1.2 1.2 Reshape Data

```

# Compute word counts for each agent's turn
df_res_field <- df_res_field %>%
  mutate(
    user_nwords = str_count(user, "\\S+"),
    ai_nwords = str_count(ai, "\\S+")
  )

df_res_long <- df_res_field %>%
  pivot_longer(
    cols = c(user_surprise, user_transience, user_nwords,
              ai_surprise, ai_transience, ai_nwords),
    names_to = c("agent", "metric"),
    names_pattern = "(user|ai)_(surprise|transience|nwords)"
  ) %>%
  pivot_wider(names_from = metric, values_from = value) %>%
  mutate(
    resonance = surprise - transience,
    agent = factor(agent, levels = c("user", "ai"), labels = c("User", "LLM"))
  )

```

1.3 1.3 Descriptive Statistics

```

desc_stats <- df_res_long %>%
  group_by(agent) %>%
  summarise(
    n = n(),
    mean_surprise = mean(surprise, na.rm = TRUE),
    sd_surprise = sd(surprise, na.rm = TRUE),
    mean_transience = mean(transience, na.rm = TRUE),
    sd_transience = sd(transience, na.rm = TRUE),
    mean_resonance = mean(resonance, na.rm = TRUE),
    sd_resonance = sd(resonance, na.rm = TRUE)
  )

print(desc_stats)

```

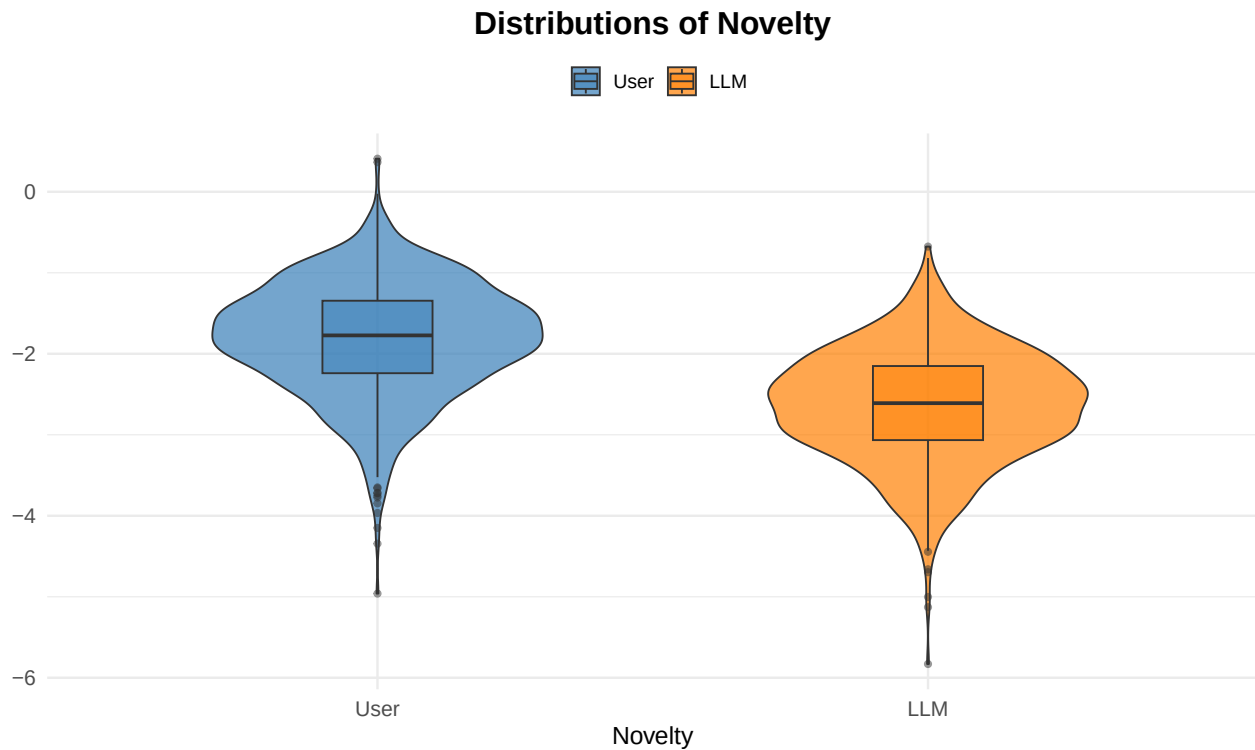
```
## # A tibble: 2 x 8
```

```
## agent      n mean_surprise sd_surprise mean_transience sd_transience
## <fct> <int>      <dbl>      <dbl>          <dbl>          <dbl>
## 1 User      789        -1.82        0.700          0.152          0.570
## 2 LLM       789        -2.65        0.696          0.143          0.625
## # i 2 more variables: mean_resonance <dbl>, sd_resonance <dbl>
knitr::kable(desc_stats, caption = "Descriptive Statistics for Novelty Metrics by Agent", digits = 3)
```

Table 1: Descriptive Statistics for Novelty Metrics by Agent

agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
User	789	-1.818	0.700	0.152	0.570	-1.970	0.870
LLM	789	-2.646	0.696	0.143	0.625	-2.762	0.887

```
p_novelty_agent <- ggplot(df_res_long, aes(x = agent, y = surprise, fill = agent)) +
  geom_violin(width = 0.6, alpha = 0.7) +
  geom_boxplot(width = 0.2, alpha = 0.5) +
  scale_fill_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00"), name = "") +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  labs(title = "Distributions of Novelty", x = "Novelty", y = NULL) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
    axis.text = element_text(size = 12)
  )
print(p_novelty_agent)
```



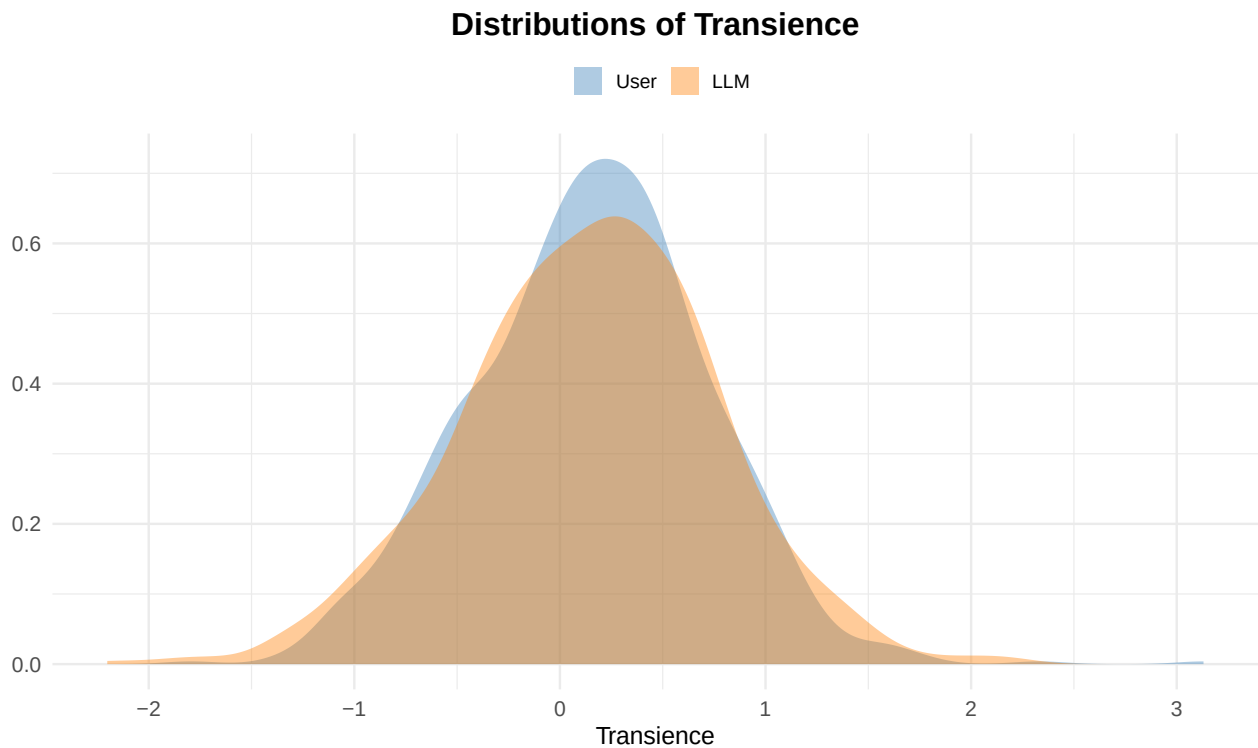
```
ggsave(
  file.path(GRAPHS_DIR, "Surprise_Distributions_Agent.png"),
  p_novelty_agent,
```

```

width = 10,
height = 6,
dpi = 300
)

p_transience_agent <- ggplot(df_res_long, aes(x = transience, fill = agent)) +
  geom_density(alpha = 0.4, color = NA) +
  scale_fill_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00"), name = "") +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  labs(title = "Distributions of Transience", x = "Transience", y = NULL) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
    axis.text = element_text(size = 12)
  )
print(p_transience_agent)

```



```

ggsave(
  file.path(GRAPHS_DIR, "Transience_Distributions_Agent.png"),
  p_transience_agent,
  width = 10,
  height = 6,
  dpi = 300
)

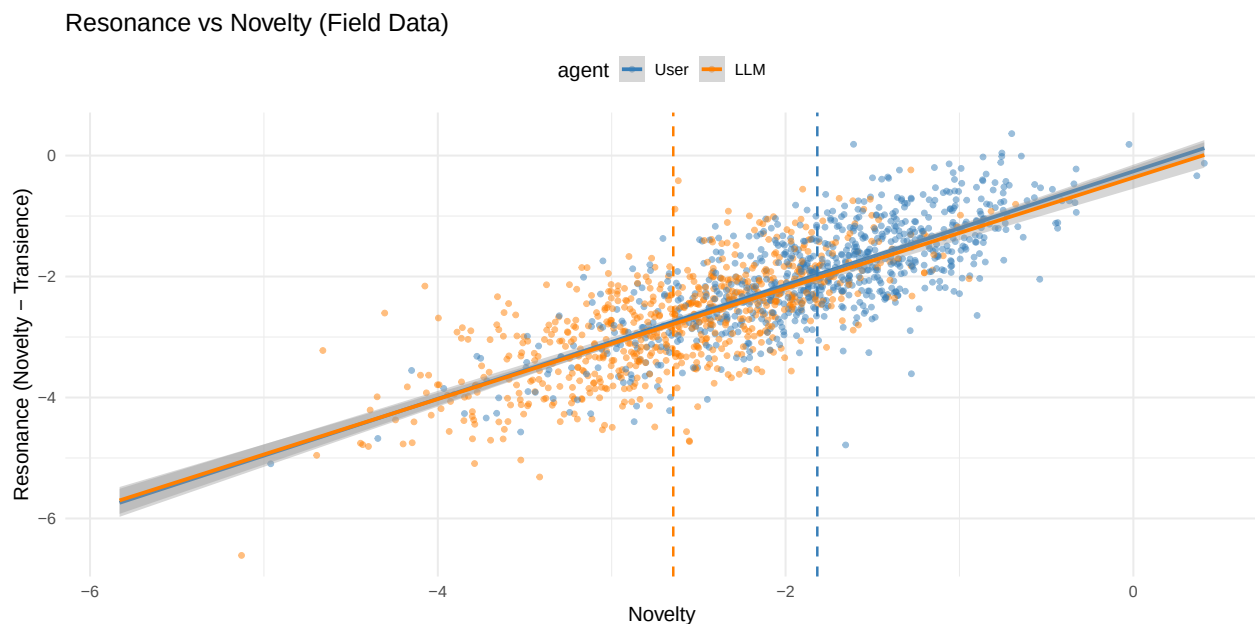
```

2. Novelty and Resonance Analysis

2.1 Resonance vs Novelty Plot

```
# Plot Resonance vs Novelty (Field Data)
p_resonance <- ggplot(df_res_long, aes(x = surprise, y = resonance, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  geom_vline(data = desc_stats,
             aes(xintercept = mean_surprise, color = agent),
             linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Resonance vs Novelty (Field Data)",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p_resonance)
```



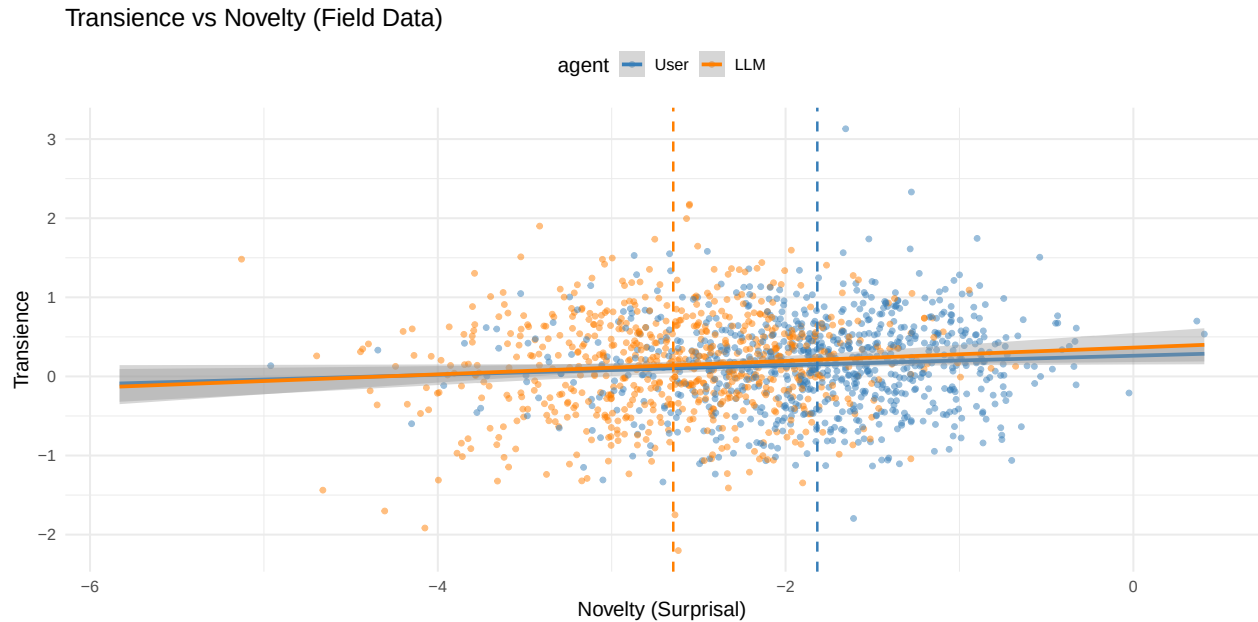
```
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_vs_novelty.png"),
        p_resonance, width = 12, height = 6, dpi = 300)
```

2.2 Transience vs Novelty Plot

```
p_transience <- ggplot(df_res_long, aes(x = surprise, y = transience, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  geom_vline(data = desc_stats,
             aes(xintercept = mean_surprise, color = agent),
             linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
```

```
labs(
  title = "Transience vs Novelty (Field Data)",
  x = "Novelty (Surprisal)",
  y = "Transience"
) +
theme_minimal(base_size = 14) +
theme(legend.position = "top")

print(p_transience)
```



```
ggsave(file.path(GRAPHS_DIR, "novelty_transience_vs_novelty.png"),
  p_transience, width = 12, height = 6, dpi = 300)
```

3. Statistical Models

3.1 Resonance Model

```
# Model without length control
model_resonance <- lmer(resonance ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Resonance Model ===\n")
```

```
## === Resonance Model ===
```

```
print(summary(model_resonance)$coefficients)
```

```
##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)  -0.20327317 0.06077152 1202.006  -3.344876  8.485940e-04
## surprise      0.97211181 0.03007574 1472.358  32.322124 1.160564e-173
## agentLLM     -0.16342864 0.10601779 1472.308  -1.541521 1.234049e-01
## surprise:agentLLM -0.05736371 0.04438793 1472.329  -1.292327 1.964468e-01
```

```
cat("\n=== Slope Comparison (Resonance) ===\n")
```

```
##
```

```
## === Slope Comparison (Resonance) ===
```

```
print(emtrends(model_resonance, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          0.972 0.0301 1472     0.913     1.03
## LLM           0.915 0.0331 1472     0.850     0.98
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

# Model WITH length control
model_resonance_len <- lmer(resonance ~ surprise * agent + nwords + (1 | conversation_id), data = df_res)
cat("\n=== Resonance Model (controlling for turn length) ===\n")

##
## === Resonance Model (controlling for turn length) ===
print(summary(model_resonance_len)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   -0.75316207 0.10466954 1017.6162 -7.195618 1.205057e-12
## surprise       0.93833635 0.03014253 1470.0383 31.129983 7.672353e-164
## agentLLM      -0.30462089 0.10687595 1468.4106 -2.850229 4.429806e-03
## nwords         0.01911518 0.00297946  819.1073  6.415652 2.369558e-10
## surprise:agentLLM -0.07654182 0.04389336 1469.6529 -1.743813 8.140075e-02
cat("\n=== Slope Comparison (Resonance, length-controlled) ===\n")

##
## === Slope Comparison (Resonance, length-controlled) ===
print(emtrends(model_resonance_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          0.938 0.0302 1470     0.879     0.998
## LLM           0.862 0.0337 1473     0.796     0.928
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

3.2 Transience Model

```
# Model without length control
model_transience <- lmer(transience ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Transience Model ===\n")

## === Transience Model ===
print(summary(model_transience)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   0.20327318 0.06077152 1202.006 3.3448759 0.0008485936
## surprise       0.02788819 0.03007574 1472.358 0.9272654 0.3539407329
## agentLLM      0.16342864 0.10601779 1472.308 1.5415209 0.1234049533
## surprise:agentLLM 0.05736371 0.04438793 1472.329 1.2923266 0.1964468573
cat("\n=== Slope Comparison (Transience) ===\n")
```

```
##
## === Slope Comparison (Transience) ===
print(emtrends(model_transience, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          0.0279 0.0301 1472  -0.0312   0.087
## LLM           0.0853 0.0331 1472   0.0203   0.150
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
# Model WITH length control
model_transience_len <- lmer(transience ~ surprise * agent + nwords + (1 | conversation_id), data = df_)

cat("\n=== Transience Model (controlling for turn length) ===\n")

##
## === Transience Model (controlling for turn length) ===
print(summary(model_transience_len)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)    0.75316207 0.10466954 1017.6162   7.195618 1.205057e-12
## surprise       0.06166365 0.03014253 1470.0383   2.045736 4.096000e-02
## agentLLM       0.30462089 0.10687595 1468.4106   2.850229 4.429806e-03
## nwords        -0.01911518 0.00297946  819.1073  -6.415652 2.369558e-10
## surprise:agentLLM 0.07654182 0.04389336 1469.6529   1.743813 8.140075e-02

cat("\n=== Slope Comparison (Transience, length-controlled) ===\n")

##
## === Slope Comparison (Transience, length-controlled) ===
print(emtrends(model_transience_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          0.0617 0.0302 1470   0.00247   0.121
## LLM           0.1382 0.0337 1473   0.07208   0.204
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

4 4. Supplementary: Resonance vs Novelty by LLM Type

```
desc_stats_llm <- df_res_long %>%
  filter(!is.na(llm_type)) %>%
  group_by(agent, llm_type) %>%
  summarise(mean_surprise = mean(surprise, na.rm = TRUE), .groups = "drop")

p_llm_user <- df_res_long %>%
  filter(!is.na(llm_type), agent == "User") %>%
  ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
  geom_point(alpha = 0.4, size = 1.2) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  geom_vline(data = desc_stats_llm %>% filter(agent == "User"),
```



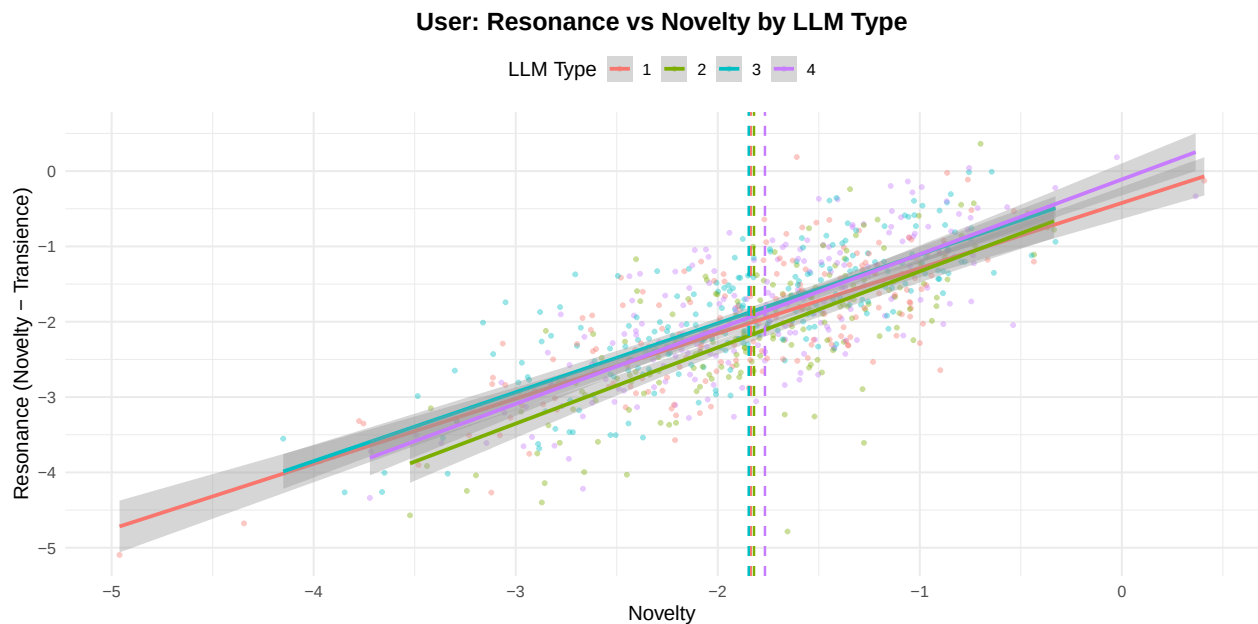
```

aes(xintercept = mean_surprise, color = factor(llm_type)),
linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
labs(
  title = "User: Resonance vs Novelty by LLM Type",
  x = "Novelty",
  y = "Resonance (Novelty - Transience)",
  color = "LLM Type"
) +
theme_minimal(base_size = 14) +
theme(legend.position = "top",
      plot.title = element_text(hjust = 0.5, face = "bold"))

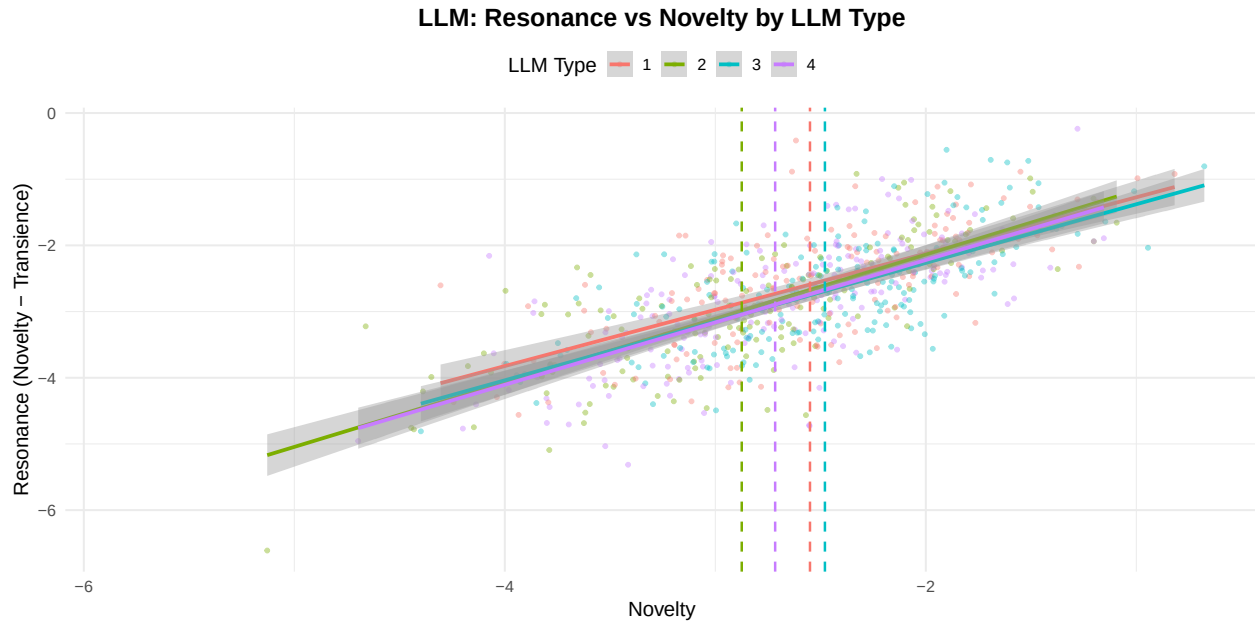
p_llm_ai <- df_res_long %>%
  filter(!is.na(llm_type), agent == "LLM") %>%
  ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
  geom_point(alpha = 0.4, size = 1.2) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  geom_vline(data = desc_stats_llm %>% filter(agent == "LLM"),
            aes(xintercept = mean_surprise, color = factor(llm_type)),
            linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "LLM: Resonance vs Novelty by LLM Type",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)",
    color = "LLM Type"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top",
        plot.title = element_text(hjust = 0.5, face = "bold"))

print(p_llm_user)

```



```
print(p_llm_ai)
```



```
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_user.png"),
  p_llm_user, width = 12, height = 6, dpi = 300)
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_ai.png"),
  p_llm_ai, width = 12, height = 6, dpi = 300)
```

5 5. Session Information

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_1.10.6 here_1.0.1 afex_1.4-1 cowplot_1.1.3
## [5] patchwork_1.3.0 zoo_1.8-12 stringi_1.8.4 arrow_19.0.1.1
```

```

## [9] lmerTest_3.1-3 lme4_1.1-35.5 Matrix_1.7-1 lubridate_1.9.3
## [13] forcats_1.0.0 stringr_1.5.1 dplyr_1.1.4 purrr_1.0.2
## [17] readr_2.1.5 tidyr_1.3.1 tibble_3.2.1 ggplot2_3.5.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1 farver_2.1.2 fastmap_1.2.0
## [4] TH.data_1.1-2 pacman_0.5.1 digest_0.6.37
## [7] timechange_0.3.0 estimability_1.5.1 lifecycle_1.0.4
## [10] survival_3.7-0 magrittr_2.0.3 compiler_4.4.2
## [13] rlang_1.1.4 tools_4.4.2 utf8_1.2.4
## [16] yaml_2.3.10 knitr_1.50 labeling_0.4.3
## [19] bit_4.0.5 plyr_1.8.9 abind_1.4-8
## [22] multcomp_1.4-26 withr_3.0.2 numDeriv_2016.8-1.1
## [25] grid_4.4.2 fansi_1.0.6 xtable_1.8-4
## [28] colorspace_2.1-1 scales_1.3.0 MASS_7.3-61
## [31] tinytex_0.52 cli_3.6.3 mvtnorm_1.3-3
## [34] rmarkdown_2.28 crayon_1.5.3 ragg_1.3.2
## [37] generics_0.1.3 rstudioapi_0.16.0 reshape2_1.4.4
## [40] tzdb_0.4.0 minqa_1.2.8 splines_4.4.2
## [43] assertthat_0.2.1 parallel_4.4.2 vctrs_0.6.5
## [46] boot_1.3-31 sandwich_3.1-1 carData_3.0-5
## [49] car_3.1-3 hms_1.1.3 pbkrtest_0.5.3
## [52] bit64_4.0.5 Formula_1.2-5 systemfonts_1.1.0
## [55] glue_1.7.0 nloptr_2.1.1 codetools_0.2-20
## [58] gtable_0.3.5 munsell_0.5.1 pillar_1.9.0
## [61] htmltools_0.5.8.1 R6_2.5.1 textshaping_0.4.0
## [64] rprojroot_2.0.4 vroom_1.6.5 evaluate_1.0.3
## [67] lattice_0.22-6 backports_1.5.0 broom_1.0.6
## [70] Rcpp_1.0.13 coda_0.19-4.1 nlme_3.1-166
## [73] mgcv_1.9-1 xfun_0.56 pkgconfig_2.0.3

```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.